

Highlights

- Stateful neural battery-state estimators run on an STM32H753ZI microcontroller.
- Thirty percent pruning cuts flash by 41–46% and inference time by 43–44%.
- Weight quantization cuts flash by 50–59% but increases mixed-precision runtime.
- Long-horizon replay shows no quantization-specific accumulation of deviation.
- Transient input faults reveal model-specific recurrent recovery behaviour.